

Reviewer Verification & Overview - What Still Needs Updating

This is the developer's original 3-page response (following pages), edited with reviewer annotations. Each issue is tagged DONE / PARTIAL / NOT DONE in the right margin, verified line-by-line against the two branches pushed on 2026-06-29: **main (Android) = 64565f0** and **Symspace_IOS_V0.1 (iOS) = 2b6c093** (the newest build). This overview covers only what is still outstanding, with the exact code to change.

Result at a glance

Confirmed DONE	Issues 2, 4, 5, 6, 7, 8 (+ Issue 1 on Android/main only)
Still outstanding	Issue 1 on iOS branch (credential logging) - HIGH
Partial	Issue 3 on iOS branch (one duplicate class remains)
Root cause	iOS branch diverged from old main; fixes not carried over

1) HIGH - iOS branch still logs login credentials (Issue #1 not finished on iOS)

On **main** the auth logs were correctly removed. But the iOS-only auth files on the newest branch still print secrets to the device log on every sign-in. These persist in Xcode Console / device logs on a release build and can be read off the phone. The exact offenders:

```
// Assets/Scripts/Home/AppleSignIn.cs (line 69)
Debug.Log("Apple IdentityToken : " + identityToken);           // <-- leaks Apple login token

// Assets/Scripts/FacebookLoginManager.cs (lines 96-97)
Debug.Log("User ID: " + AccessToken.CurrentAccessToken.UserId);
Debug.Log("Access Token: " + AccessToken.CurrentAccessToken.TokenString); // <-- leaks FB token
```

Fix - Option A (simplest): delete the three lines. They are development-only and add no production value.

Fix - Option B (keep for debugging, but strip from release): route them through a conditional wrapper so the compiler removes them from shipping builds:

```
// New file: Assets/Scripts/Utils/SymspaceDebug.cs
public static class SymspaceDebug {
    [System.Diagnostics.Conditional("UNITY_EDITOR")]
    [System.Diagnostics.Conditional("DEVELOPMENT_BUILD")]
    public static void Log(string message) => UnityEngine.Debug.Log(message);
}

// Then replace the logs above with a NON-sensitive message, e.g.:
SymspaceDebug.Log("Apple sign-in success"); // stripped from release builds, zero overhead
```

2) MEDIUM - Duplicate ErrorResponse still on iOS (Issue #3 partial)

main has exactly one `ErrorResponse` class. The iOS branch still has two (`FirebaseAuthManager.cs` and `FacebookLoginManager.cs`). Remove the second so error handling is consistent:

```
// In Assets/Scripts/FacebookLoginManager.cs -- DELETE this nested/duplicate block:
[System.Serializable]
public class ErrorResponse { public string message; /* ... */ }

// Keep the single shared definition in FirebaseAuthManager.cs and add, if needed:
// using
```

3) ROOT CAUSE - the two branches have diverged (fix this to stop repeats)

The iOS branch was created from the **old 2026-06-19 main** (shared ancestor commit 7f4bec4), not the updated 2026-06-29 main. That is why main is clean while iOS still has the token logs and the duplicate class - the Android fixes never reached iOS. Reconcile before doing more work, so fixes stop having to be done twice:

```
# Bring the 2026-06-29 Android fixes into the iOS branch
git checkout Symospace_IOS_V0.1
git merge main          # or: git rebase main
# resolve conflicts, then re-verify no credential logs remain:
git grep -nE "Debug\.Log.*(Token|Access|Email)" -- Assets/Scripts
```

4) Minor - confirm the stray scene

The iOS build list contains `Hand Tracking.unity`; there is also a `Hand Tracking 1.unity` in the project. Confirm which one ships and delete the other to avoid loading the wrong scene.

Everything else (Issues 2, 4, 5, 6, 7, 8) is verified complete - see the per-issue tags on the developer's pages that follow.

Response to Code Review Verification & Findings

This document addresses the findings in the verification review and describes the updates made after that review was completed.

Issue #1: Security Logging / Debug Wrapper

PARTIAL - iOS FAIL

The authentication debug logs mentioned in the audit were temporary logs used during development and testing.

During the initial cleanup, these logs were commented out. Based on the review, all remaining authentication and development-only logs have now been completely removed instead of being left as comments.

I also completed a final review to ensure that production builds no longer include unnecessary authentication or user-related debug logs.

Issue #2: GC / Allocation Optimization in Hand Tracking

DONE

The audit correctly identified that the memory optimization applied to the ring tracking system had not yet been added to the watch/wrist tracking implementation.

This has now been updated. The same zero-allocation optimization has been applied to both tracking systems, including the HandTrackingVisualizer, where the per-frame `buffer.ToArray()` allocation previously existed.

This removes unnecessary memory allocations and helps reduce garbage collection spikes and frame drops during AR tracking.

Issue #3: Duplicate ErrorResponse Classes

DONE / iOS partial

The report correctly identified that multiple `ErrorResponse` class definitions still existed in the project.

During the initial cleanup, only some duplicate classes were removed, while others were missed.

I have now completed a full review of the project and removed all remaining duplicate definitions. The project now uses a single shared `ErrorResponse` class across all services and screens.

This keeps error handling consistent throughout the application and avoids different implementations in different modules.

Issue #4: Per-frame Logging in BodySpawned Script

DONE

The audit identified a BodySpawned script that contained a debug log which could affect performance if it was used in an active scene.

After reviewing the project, I confirmed that this script was not referenced in any active scene or production workflow. It only contained an empty Update() method and a single debug log statement.

Since it was no longer used, the script has been removed from the project.

Issue #5: Missing AR Scenes in Build Settings

DONE

The missing scenes reported in the audit are related to AR hand tracking and body tracking features that are available only in the iOS version of the application.

The repository reviewed during the audit was the Android project branch, which intentionally does not include the iOS-specific AR tracking scenes in its build settings.

To separate platform-specific features more clearly, I have created and uploaded a dedicated iOS branch that includes:

- Hand Tracking scene
- Body Tracking scene
- ARKit and other iOS-specific tracking workflows

Since these features depend on iOS AR capabilities, they are not expected to be available in the Android branch.

Issue #6: Apple Sign-In Wiring

DONE

The SingInWithApple typo was already corrected to SignInWithApple.

Apple Sign-In is implemented only for the iOS version of the application and is not included in the Android branch that was reviewed.

For that reason, the related UI flow and serialized button bindings are not expected to appear in the Android scenes.

The dedicated iOS branch containing the Apple Sign-In implementation has also been uploaded together with the iOS AR tracking features.

Issue #7: Category Parsing Edge Case

DONE

The audit identified an edge case where an invalid parsing result could still overwrite the selected category.

The logic has now been updated so that the category is changed only when parsing is successful.

This prevents invalid input from replacing the current selection and makes the behavior more reliable.

Issue #8: Repository Cleanup

DONE

The report identified Unity recovery files and temporary scene files that had been committed to the repository.

After reviewing the repository, I confirmed that these were editor-generated files and not part of the actual project.

These files have now been removed as part of the repository cleanup, keeping the project cleaner and reducing unnecessary files in version control.